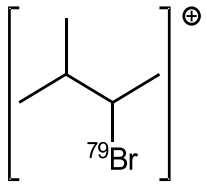
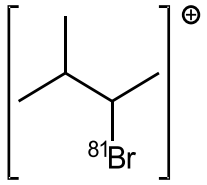
4(a)(i)	**M1** (molecules with the) same molecular formula / same number of atoms of each element **M2** different structural formulae/structures	**2**

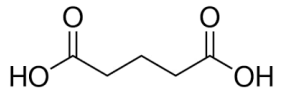
4(a)(v)	**M1** (reaction proceeds via a) more stable intermediate / carbocation **M2** because of greater positive inductive effect of more alkyl groups	**2**

4(a)(vi)	**M1**  **M2** 	**2**

Examiner report missing
9701_w25_qp_24 Question 4

1(a)(i)	24 protons 28 neutrons 24 electrons	**1**
1(a)(iii)	(relative) abundance (of each isotope)	**1**
1(a)(iv)	The fourth isotope has any value of RIM less than 51.996	**1**

Examiner report missing
9701_w25_qp_23 Question 1

3(a)	simplest /lowest (whole number / integer) ratio of atoms (of each element) in a compound / molecule	**1**
3(e)(i)		**1**
3(e)(ii)	**M1** absorption labelled at approx. 3000 cm^{-1} identified as O—H **M2** absorption labelled at approx. 1720 cm^{-1} identified as C=O	**2**

Examiner report missing
9701_w25_qp_23 Question 3

4(a)(i)	reduction	**1**
4(c)(i)	2,3-dimethylbut-1-ene	**1**
4(c)(ii)	(The spectrum is of) **X** **AND** 41 is $[CH_2=CCH_3]^+$ / 41 is $C_3H_5^+$ **OR** 43 is $[CH_3CHCH_3]^+$ / 43 is $C_3H_7^+$ **OR X** would NOT have significant peak at 54 / $[(CH_3)_2C=C^+]$ / $C_4H_6^+$ **OR X** would NOT have significant peak at 42 due to $[(CH_3)_2C^+]$ / $C_3H_6^+$	**1**
4(c)(iii)	($[M+1]^+$ / 34.7 × 100 / 1.1 = 6) $[M+1]^+ = 2.29$	**1**

Examiner report missing
9701_w25_qp_23 Question 4

1(a)(i)	25 protons 30 neutrons 25 electrons	**1**
1(a)(ii)	(atoms of an element) with the same number of protons and different number of neutrons	**1**
1(a)(iii)	**M1** (relative) isotopic mass. **OR** (relative isotopic) mass of Mn-55 and Mn-53 **M2** (relative) abundance of each isotope	**2**

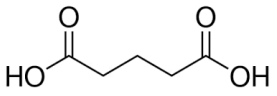
Examiner report missing
9701_w25_qp_22 Question 1

4(a)(i)	free-radical substitution	**1**
4(d)(i)	condensation	**1**
4(d)(ii)	**M1** absorption labelled at approx. 1750 cm^{-1} identified as C=O **M2** absorption labelled at approx. 1650 cm^{-1} identified as C=C **M3** absorption labelled at approx. 1170 cm^{-1} identified as C—O	**3**
4(d)(iii)	(0.31 / 4.7 × 100 / 1.1) = 6 carbon atoms	**1**

Examiner report missing
9701_w25_qp_22 Question 4

1(a)(i)	24 protons 28 neutrons 24 electrons	**1**
1(a)(iii)	(relative) abundance (of each isotope)	**1**
1(a)(iv)	The fourth isotope has any value of RIM less than 51.996	**1**

Examiner report missing
9701_w25_qp_21 Question 1

3(a)	simplest /lowest (whole number / integer) ratio of atoms (of each element) in a compound / molecule	**1**
3(e)(i)		**1**
3(e)(ii)	**M1** absorption labelled at approx. 3000 cm^{-1} identified as O—H **M2** absorption labelled at approx. 1720 cm^{-1} identified as C=O	**2**

Examiner report missing
9701_w25_qp_21 Question 3

4(a)(i)	reduction	**1**
4(c)(i)	2,3-dimethylbut-1-ene	**1**
4(c)(ii)	(The spectrum is of) **X** **AND** 41 is $[CH_2=CCH_3]^+$ / 41 is $C_3H_5^+$ **OR** 43 is $[CH_3CHCH_3]^+$ / 43 is $C_3H_7^+$ **OR X** would NOT have significant peak at 54 / $[(CH_3)_2C=C^+]$ / $C_4H_6^+$ **OR X** would NOT have significant peak at 42 due to $[(CH_3)_2C^+]$ / $C_3H_6^+$	**1**
4(c)(iii)	($[M+1]^+$ / 34.7 × 100 / 1.1 = 6) $[M+1]^+ = 2.29$	**1**

Examiner report missing
9701_w25_qp_21 Question 4

6(a)(i)	simplest (whole number) ratio of atoms (of the elements) in a compound	**1**
6(c)(i)	**M1** $\frac{100}{1.1} \times \frac{11}{100}$ **M2** 10	**2**
6(c)(ii)	$C_{10}H_{18}$	**1**
6(c)(iii)	$(C_4H_9)^+$	**1**

Examiner report missing
9701_s25_qp_24 Question 6

5(a)(i)	C_2H_3O	**1**

5(b)(i)	number of σ bonds = 14 number of π bonds = 2	**2**

5(b)(v)	• **A** C–H alkane • **B** C=O ester • **C** C=C alkene • **D** C–O ester 2 points = 1 mark, all 4 points = 2 marks	**2**

Examiner report missing
9701_s25_qp_23 Question 5

2(a)(i)	**route 1** **M1** average (weighted) mass of (an) atom(s) (of an element) **OR** average (weighted) mass of the isotope(s) (of an element) **M2** compared to the mass of the unified atomic mass unit **OR** on a scale in which a carbon-12 atom / isotope has a mass of exactly 12 (units) **OR** on a scale in which the mass of a carbon-12 atom/isotope is exactly 12 (units) **OR** compared to $\frac{1}{12}$ mass of one atom of C-12 **route 2** **M1** average mass of one mol of isotopes / atoms (of an element) **M2** relative / compared to $\frac{1}{12}$ of the mass of one mol of C-12 atoms/isotopes **OR** when the mass of one mol C-12 atoms / isotopes is exactly 12 (000) g	**2**

2(a)(ii)	**M1** relative abundance of unknown isotope = (100 – 91.9 – 6.0) = 2.1% **M2** 55.8 = (53.9 × 6.0 + 55.9 × 91.9 + ? × M1) / 100 = 56.9	**2**

Examiner report missing
9701_s25_qp_22 Question 2

5(a)	$C_3H_4O_3$	**1**

5(b)(i)	$5.00 \times 10^{-3} \times 36.65 / 1000$ $= 1.833 \times 10^{-4} / 1.8325 \times 10^{-4}$	**1**

5(b)(iii) **1**

absorption / cm^{-1}	present in spectrum of vitamin C	present in spectrum of **M**
1500–1680	✓	× / – / blank
2850–2950	✓	✓
3200–3650	✓	✓

Examiner report missing
9701_s25_qp_22 Question 5

2(a)	atoms (of an element) with same number of protons and different number of neutrons	**1**

2(b)(i)	**M1** method showing each abundance multiplied by m/e value • (5.9×54) and (91.9×56) and (2.2×57) **OR** • (0.59×54) and (9.19×56) and (0.22×57) **M2** the three values from **M1** expressed as a percentage and added together correctly with final answer to 1 d.p $\frac{5.9}{100} \times 54 + \frac{91.9}{100} \times 56 + \frac{2.2}{100} \times 57 = 55.9$	**2**

Examiner report missing
9701_s25_qp_21 Question 2

<table>
<tr><td>4(a)</td><td>$C_3H_4O_3$</td><td>1</td></tr>
<tr><td>4(b)(i)</td><td>$(28.40 / 1000 \times 0.0005) = 1.42 \times 10^{-5}$ mol</td><td>1</td></tr>
<tr><td>4(c)</td><td>Any two complete rows from:
<table><tr><th>wavenumber</th><th>bond</th><th>functional group</th></tr><tr><td>1040 – 1300</td><td>C–O</td><td>ester / hydroxy</td></tr><tr><td>3200 – 3650</td><td>O–H</td><td>hydroxy / alcohols</td></tr><tr><td>2850 – 2950</td><td>C–H</td><td>alkane</td></tr><tr><td>1710 – 1750</td><td>C=O</td><td>ester</td></tr></table></td><td>2</td></tr>
</table>

Examiner report missing
9701_s25_qp_21 Question 4

1(a)(i)	• (+)3 / III • –1 / I • (+)5 / V • –1 / I	**2**
1(c)(ii)	ethanol (solvent) **AND** heat	**1**
1(d)(i)	(relative) abundance is 3:1 / 100:33 **AND** link to two isotopes of C*l*	**1**
1(d)(ii)	**D** is C*l*—C≡C—H	**1**

Examiner report missing
9701_m25_qp_22 Question 1

1(a)(i)	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$	**1**

1(c)	mass of an atom of an isotope	**1**
	compared to the mass of the unified atomic mass unit (amu) OR on a scale on which a carbon-12 atom has a mass of exactly 12 units OR divided by $\frac{1}{12}$ mass of a carbon-12 atom	**1**

1(c)(ii)	$\frac{190.96 \times 15.18 + 192.96 \times 34.82}{50.00}$	**1**
	= **192.35**	**1**

Question 1

(a) (i) Most candidates were able to give a correct electronic configuration of: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$. A few candidates included the $3d^74s^2$ given in the question. This was not required in the answer.

(ii) This question was well answered with most drawings attempting to represent a sphere as a circle. A few diagrams showed the x, y and z axes drawn to represent a 3D shape.

(iii) Several candidates gave an answer of one, whilst many gave the correct answer.

(b) Most candidate answers were correct for the total number of protons and neutrons, but the total number of electron shells, for rhodium, were often incorrectly given as 4.

(c) (i) The definition was not well answered. Many candidates stated that it was the average mass of an isotope/weighted average mass of an isotope or the average mass of isotopes. Very few answers gave a complete definition.

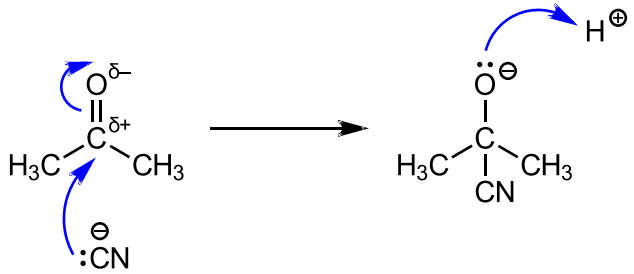
(ii) Many candidates incorrectly used all three isotopes to calculate the A_r of iridium and divided by 100 or used the two isotopes of iridium but divided by 100 instead of by 50.

(d) (i) Many candidates were aware that to calculate an integer value of x, the water of crystallisation in $RhCl_3 \cdot xH_2O$, a mole ratio calculation was required. The ratio of H_2O to $RhCl_3$ was given/deduced as 20.5 per cent : 79.5 per cent by mass, from which a mole ratio of 0.38 : 1.14 can be calculated. Many candidates chose a slightly different route and used the expression $18x \div (209.4 + 18x) = 0.205$, to correctly calculate a vale of $x = 3$.

(ii) Many candidate answers mentioned that stereoisomers contained the same 'molecular formula' but omitted the fact that they also contained the same 'structural formula', which is the key feature for this definition. Several candidates correctly stated that the stereoisomers had 'different arrangements in space' but made no mention of what was arranged differently in space e.g. atoms/groups.

(iii) Many candidates failed to recognise that a single product would be formed and gave explanations about adding extra CH_2 units to the chain.

iv) The diagrams of the two stereoisomers of but-2-ene were often correct. Errors included confusion on the position of the C=C, in the chain, on one or both structures and incorrect groups or incorrect arrangements.

4(a)(i)	 • curly arrow from lone pair on C of CN^- to carbonyl C • correct dipole on C=O • curly arrow from C=O bond to O • correct intermediate (the lone pair(s) not needed for this bullet point) • curly arrow from lone pair on —O to (source of) H^+	**1**
		1
		1

4(f)(i)	All three have a C—H **OR** CH **bond**	**1**

4(f)(ii)	compound **A** **AND** absorption at 2200–2250 cm^{-1} indicates C≡N	**1**

Question 4

(a) (i) The mechanism attracted many good answers. The common errors included: arrows from the negative charge (on the CN^- ion) instead of from the lone pair on the C atom, missing polarity, specifically missing the $\delta+$ on the H (of H-CN), and incorrect attack of the intermediate by the CN^- ion.

(ii) Many answers stated that there were two CH_3 groups on one C atom in A but did not go further to explain the significance of this in terms of the lack of optical isomerism in A. Credit was given for stating that no chiral centre was present or that the central C atom was not bonded to four different atoms/groups.

(b) Candidates often stated that the mineral acids HCl or H_2SO_4 were required for the hydrolysis reaction of a CN group to a carboxylic acid but missed the requirement for the acid to be dilute or aqueous.

(c) (i) The equation was frequently correct. A common error was the omission of 2(H), from the equation, or using just 2H, without the brackets.

(ii) The name of compound C was well known.

(d) Most candidate answers stated that an orange precipitate was formed when propanone was reacted with 2,4-DNPH.

(e) A common answer was that 'Fehling's reagent only reacts with aldehydes', with no reference to any chemistry in this reaction. Fehling's reagent behaves as an oxidising reagent and ketones are not easily oxidised, hence, there is no reaction between Fehling's reagent and propanone.

(f) (i) Many candidate answers stated that the absorptions at 2850 – 2950 cm^{-1} were attributable to a C-H bond. As all three compounds in the question contained groups containing C-H bonds these compounds were indistinguishable using these absorptions.

(ii) Many candidates correctly identified the absorption at 2200 – 2250cm^{-1}, which was unique to compound A, as it was the only compound containing a C≡N bond. As all three compounds, A, B and C contained an O-H bond, no credit was given for frequent mention of this absorption.

1(a)(i)	lowest energy state owtte examples • (with) low**est** energy • (with) no external energy • (with) no added energy • (where) electrons are not promoted to a higher energy level • most energetically stable	**1**

1(b)(i)	<table><tr><th>isotope</th><th>RIM</th><th># protons</th><th># neutrons</th></tr><tr><td>$^{93}_{41}$Nb</td><td>92.91</td><td>● 41</td><td>● 52</td></tr><tr><td>$^{181}_{73}$Ta</td><td>180.95</td><td>● 73</td><td>● 108</td></tr></table>	**2**

1(b)(iii)	92.91× 0.909 + 180.95 × 0.091	**1**
	= 100.92	**1**

Question 1

(a) (i) Understanding of 'ground state' was rarely described correctly. Confusion of 'ground state' with 'natural' or 'standard' state was seen.

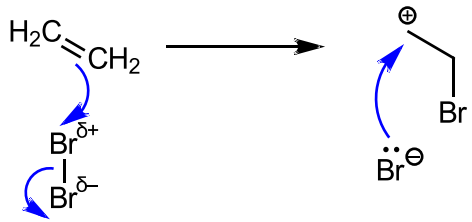
(ii) Many candidates gave the correct representation of the arrangement of electrons in the 3d and 4s orbitals.

(iii) The most common incorrect answer was '6'; perhaps deduced by those who considered only one of the energy levels n = 2 or n = 3.

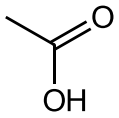
(b) (i) Most candidates gave the correct deduction of the number of protons and neutrons in each species.

(ii) Confusion of the terms 'relative atomic mass' with 'relative isotopic mass' was seen in weaker candidate responses. Comparisons of mass were often incomplete or incorrect, examples included no identification of the specific carbon isotope.

(iii) Fully correct answers used the relative isotopic mass given in Table 1.1 correctly and gave their final answer to 2 decimal places. Weaker responses did not select the relevant data from Table 1.1. Final answers were not always given to the 2 decimal places specified.

4(a)	 **pt 1** dipole on Br_2 **pt 2** curly arrow from C=C bond to a $Br^{\delta+}$ **pt 3** curly arrow from Br–Br bond to the other Br **pt 4** correct intermediate **pt 5** curly arrow from lone pair on Br^- to C^+ **5•** = 3 marks. **4•** = 2 marks. **3•** = 2 marks. **2•** = 1 marks. **1•** = 0 marks.	**3**

4(d)(i)	structural / positional	**1**

4(e)(i)	**M1** identify **H** =  **M2** (broad) absorption within the range 3600–2500 cm^{-1} so O-H (bond) O-H (bond) is equivalent to OH bond **M3** pt 1 and pt 2 •✓ **OR** pt 1 and pt 3 •✓ •**pt 1** absorption within the range 1670–1750 cm^{-1} so C=O (bond) •**pt 2** absorption within the range 1040–1300 cm^{-1} so C-O (bond) •**pt 3** (M^+ at m/e = 60 so it has) molecular mass / Mr = 60	

Question 4

(a) Well-rehearsed candidate answers included the accurate placement of curly arrows, dipoles and the correct intermediate. Weaker answers often had ambiguous placement of curly arrows, inappropriate use of partial charges, incorrect dipole on Br_2 and incorrect intermediates.

(b) Correct calculations, sometimes with an appropriate Hess' cycle in their working, were seen. Incorrect answers were calculated by those candidates that did not fully understand or use the definition of enthalpy change of formation when applying Hess's law.

(c) (i) The product **A** was correctly shown in many answers. Candidates need to show correct connectivity of the O-H group to the carbon backbone when structures are drawn.
The naming of **B** presented problems for some candidates. Incorrect names often described the incorrect position of Br on the carbon backbone or gave a name which described the wrong number of carbon atoms in the backbone.

(ii) Unambiguous structures were seen in a significant number of candidate answers.

(iii) This question proved to be extremely demanding. All the information provided in the details of Fig. 5.1 needed to be considered, including the three products as well as the reaction conditions.

(d) (i) Correct identification of the type of isomerism was given in a significant number of candidate answers. Chain isomerism was a common incorrect answer even though the two molecules were made up of only two carbon atoms.

(ii) Many candidate answers stated that **F** was a carbonyl compound; this term describes the functional group present in the molecule and is common to both homologous series; aldehyde and ketone.

(iii) Many candidate answers gave appropriate observations for the behaviour of **F** with 2,4-DNPH reagent and Tollen's reagent. A much smaller number of answers identified a positive reaction with appropriate observation for the reaction with alkaline I_2(aq).

(e) (i) This question proved to be demanding. The strongest candidates gave full explanations of their interpretation of the infrared spectrum by relating relevant wavenumber values at absorption troughs to appropriate bond types and used the molecular ion peak to deduce the molecular mass of **H**. Weaker responses quoted key wavenumber values at troughs in the infrared spectrum and restated that the molecular ion peak in the mass spectrum was at m/e = 60 without explaining what this information actually provides about structure **H**.

(ii) The correct role of reagent **G** was given in many candidate answers. Catalyst was a common incorrect answer.

1(a)(i)	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$	**1**

1(c)	mass of an atom of an isotope	**1**
	compared to the mass of the unified atomic mass unit (amu) OR on a scale on which a carbon-12 atom has a mass of exactly 12 units OR divided by $\frac{1}{12}$ mass of a carbon-12 atom	**1**

1(c)(ii)	$\frac{190.96 \times 15.18 + 192.96 \times 34.82}{50.00}$	**1**
	= **192.35**	**1**

Question 1

(a) (i) Most candidates were able to give a correct electronic configuration of: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$. A few candidates included the $3d^74s^2$ given in the question. This was not required in the answer.

(ii) This question was well answered with most drawings attempting to represent a sphere as a circle. A few diagrams showed the x, y and z axes drawn to represent a 3D shape.

(iii) Several candidates gave an answer of one, whilst many gave the correct answer.

(b) Most candidate answers were correct for the total number of protons and neutrons, but the total number of electron shells, for rhodium, were often incorrectly given as 4.

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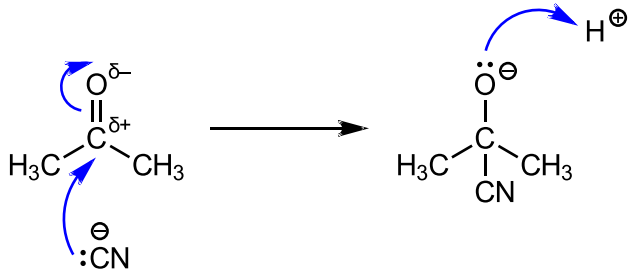
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(iii) Many candidates failed to recognise that a single product would be formed and gave explanations about adding extra CH_2 units to the chain.

iv) The diagrams of the two stereoisomers of but-2-ene were often correct. Errors included confusion on the position of the C=C, in the chain, on one or both structures and incorrect groups or incorrect arrangements.

4(a)(i)	 • curly arrow from lone pair on C of CN^- to carbonyl C • correct dipole on C=O • curly arrow from C=O bond to O • correct intermediate (the lone pair(s) not needed for this bullet point) • curly arrow from lone pair on —O to (source of) H^+	**1**
		1
		1

4(f)(i)	All three have a C—H **OR** CH **bond**	**1**

4(f)(ii)	compound **A** **AND** absorption at 2200–2250 cm^{-1} indicates C≡N	**1**

Question 4

(a) (i) The mechanism attracted many good answers. The common errors included: arrows from the negative charge (on the CN^- ion) instead of from the lone pair on the C atom, missing polarity, specifically missing the $\delta+$ on the H (of H-CN), and incorrect attack of the intermediate by the CN^- ion.

(ii) Many answers stated that there were two CH_3 groups on one C atom in A but did not go further to explain the significance of this in terms of the lack of optical isomerism in A. Credit was given for stating that no chiral centre was present or that the central C atom was not bonded to four different atoms/groups.

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(c) (i) The equation was frequently correct. A common error was the omission of 2(H), from the equation, or using just 2H, without the brackets.

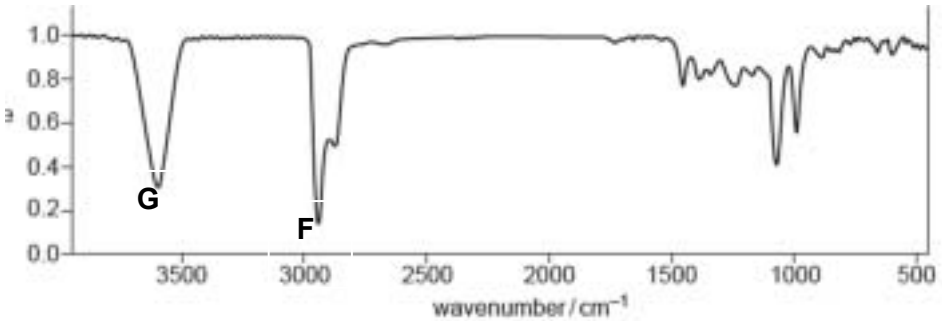
(ii) The name of compound C was well known.

(d) Most candidate answers stated that an orange precipitate was formed when propanone was reacted with 2,4-DNPH.

(e) A common answer was that 'Fehling's reagent only reacts with aldehydes', with no reference to any chemistry in this reaction. Fehling's reagent behaves as an oxidising reagent and ketones are not easily oxidised, hence, there is no reaction between Fehling's reagent and propanone.

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(ii) Many candidates correctly identified the absorption at 2200 – 2250cm^{-1}, which was unique to compound A, as it was the only compound containing a C≡N bond. As all three compounds, A, B and C contained an O-H bond, no credit was given for frequent mention of this absorption.

5(a)	cold dilute $KMnO_4$ **OR** cold dilute potassium manganate(VII)	**1**
5(b)(i)	step 1 …..Br_2 (in the dark) step 2 …..NaOH(aq) + heat	**2**
5(c)	 **M1** peak shown around 2850–2950 labelled F **AND** (corresponds to) C—H **M2** peak shown around 3200–3600 labelled **G AND** (corresponds to) O—H (in hydroxy(l) / alcohol)	**2**

Question 5

This question tested organic reactions and spectroscopic methods using cyclohexene as a starting material. Many candidates seemed familiar with the chemistry, although the context of a cyclic molecule proved to be disconcerting for a few, who appeared to misinterpret the information given, or did not concentrate solely on the relevant functional groups.

(a) This was well answered by many, although differences appeared in the conditions to use with the $KMnO_4$ reagent (e.g. concentrated/dilute, alkaline/neutral/acidic).

(b)(i) Step 1 was familiar to many candidates. The hydrolysis of two functional groups in step 2 led many to think that H_2O_2 was being used, rather than two portions of NaOH.

(ii) In asking for a mechanism, 'electrophilic' was expected as well as 'addition'.

(c) Candidates showed some understanding in answering this question but often missed full credit by not following the rubric properly: the diagram needed labelling and each responsible bond (stretch) needed to be clearly identified. Many candidates only achieved half of this, or switched **F** and **G**.

(d)(i) This was mostly well answered, although some candidates lost credit when opting to give fully displayed versions of the CN group, neglecting to show a triple C≡N bond or using the wrong connectivity (N attached to the ring).

(ii) This was answered well by many candidates. Others did not acknowledge that acidic conditions are used, creating an ammonium species rather than NH_3.

(iii) The formulae of the organic products were generally well indicated, though candidates often then misbalanced the equation with insufficient (H).

(iv) This was generally well answered by candidates. A few $NaBH_4$ responses were offered, to no credit: carboxylic acids need the stronger of the two hydride reagents on the syllabus.

Question 5

(a) (i) Those candidates with a good understanding of mass spectrometry realised that the peaks shown in Table 5.1 represented the molecular ion peak, M^+, and its $(M+1)^+$ peak and used the appropriate formula to calculate the relative abundance of the M+1 peak. Some weaker responses based their calculation on an incorrect formula.

(ii) The strongest responses engaged with the information about compound **W**. Weaker responses did not engage fully with the question details and answered in terms of structural formulae. Some formulae gave the fragments with no charges or negative charges.

(b) (i) An understanding that an O–H bond, due to a carboxyl functional group, produces a broad stretch within the range 2500–3000 cm^{-1} was seen in the best responses. The weakest responses did not engage with the details given for compound **W** and its reaction to produce **X**, and listed every potential functional group which might account for bond **A** and bond **B**, including amide and ester functional groups.

(ii) Of those candidates that correctly identified **B** as a C=O of a carbonyl group, some candidates ignored the reaction conditions stated in the question and represented **X** as an aldehyde rather than a ketone.

(c) (i) A significant proportion of candidates correctly named the functional group present in **Y**, based on the information provided.

(ii) Many equations were shown with the correct formula of the organic product; not all equations were balanced. Identification of H_2O rather than H_2 as the other product was seen.

(iii) Those candidates who applied knowledge of the behaviour of different types of alcohol and carbonyl compounds correctly deduced the structure of **Y**. Confusion about the chemical reactions of alcohols and carbonyl compounds was seen in weaker responses.

6(a)(i)	(molecules with the) same structural formula (and same molecular formula) with different arrangement of atoms / groups in space	**1**
6(c)(i)	**M1** deduce n = 12 (from **Y** – 1C (in methanol)) **M2** $(100 \times x) \div (1.1 \times 100) = 12$ (so) $x = 13.2$	**2**
6(c)(ii)	<table><tr><th></th><th>observation on addition of 2,4–DNPH</th></tr><tr><td>**Y**</td><td>orange precipitate</td></tr><tr><td>**Q**</td><td>no precipitate</td></tr></table> Both correct for one mark	**1**
6(c)(iv)	**M1 Y** will have absorption / peak / trough between 1670–1740 due to C=O (**Q** will not) **M2 Q** will have absorption / peak / trough between 3200–3600 due to O-H (**Y** will not)	**2**

Question 6

(a) (i) The definition of stereoisomerism was one of the least well answered questions. The most common error was to state that 'molecules had the same molecular formula but different structural formula'. Few mentioned that it was the arrangement of the atoms or groups in space.

(ii) Very few answers gained full credit here. Some candidates identified a C=C double bond and two chiral centres and then incorrectly stated that this gave (2 × 3 =) 6 stereoisomers instead of using the correct expression 2^3 (= 8).

(iii) The molecular formula of **V** was answered quite well. There were no common errors.

(iv) Many candidates correctly identified an alkene and carbonyl group. The most common error was to omit the ester group and substitute it for an alcohol group or a carboxyl/carboxylic acid group.

(b) (i) The question required an identification of the role of reagent **T**. Candidates who stated that **T** behaved as a reducing agent gained full credit. Those candidates who ventured to explain the behaviour of **T** with each functional group generally understood that it reduced the carbonyl group in **V** to an alcohol group in **W**. The reaction of the C=C to from a saturated chain attracted a variety of incorrect answers ranging from acting as a catalyst to hydrolysing the alkene group.

(ii) This question was well answered.

(c) (i) Candidates found this very challenging. Many answers used the m/e values of 200 and 201, from the mass spectrum, instead of the relative abundance values of 100 and x, and attempted to average these values to determine the relative atomic mass.

Creditworthy answers deduced that **Q** contained 12 C atoms and then used this value of *n* in the expression $n = \dfrac{100 \times \text{abundance of M} + 1 \text{ ion}}{1.1 \times \text{abundance of M} + \text{ion}}$

to calculate the abundance of the (M + 1) ion which is equivalent to the value of x.

(ii) This question was answered correctly by many candidates with the most common error being to omit 'precipitate' with the orange colour for the observation with 2,4-DNPH with **Y**.

(iii) The majority of candidates did not recognise that the volume of hydrogen gas was collected at s.t.p., in the question, which, from the Table of data, is 22.4 $dm^3 mol^{-1}$.

Various methods were used to calculate the moles of gas produced to include the ideal gas equation and 24.0 $dm^3 mol^{-1}$.

Those candidates who managed to calculate the number of moles of gas then needed to appreciate that the ratio of hydroxyl groups to hydrogen gas was 2 : 1. The award of full credit was rare.

(iv) Many candidates gave correct absorption ranges for one of the two carbonyl absorptions in **Y**, for the ketone and ester functional groups. Several answers gave a range of absorptions for a carboxyl group in **Q** as a difference in the infrared spectrum, having not appreciated that **Q** contains **only** hydroxyl groups, mentioned in the stem of **(c)(ii)**.

4(a)	<table><tr><td>molecule</td><td>**P**</td><td>**Q**</td><td>**R**</td><td>**S**</td></tr><tr><td>geometrical</td><td>✓</td><td>(×)</td><td>(×)</td><td>(×)</td></tr><tr><td>optical</td><td>(×)</td><td>✓</td><td>✓</td><td>✓</td></tr></table>	**1**
	<table><tr><td>molecule</td><td>**P**</td><td>**Q**</td><td>**R**</td><td>**S**</td></tr><tr><td>geometrical</td><td>✓</td><td>(×)</td><td>(×)</td><td>(×)</td></tr><tr><td>optical</td><td>(×)</td><td>✓</td><td>✓</td><td>✓</td></tr></table> $CH_3CHBrCH_2CH_3$	**1**

4(c)(i)	n = 100 / 1.1 × 0.55 / 7.2	**1**
	= 6.94 (so 7 carbon atoms)	**1**

4(c)(ii)	FG = ester / (R)COOR(')	**1**
	(absorption at) 1040–1300 cm^{-1} indicates C—O (stretch) **AND** (absorption at) 1710–1750 cm^{-1} indicates (C=O stretch)	**1**

Question 4

This an organic and spectroscopic question. Candidates are advised to be clear in matters of structure, both when representing molecules diagrammatically and when referring to features within a molecule, such as those identified by different types of spectra.

(a) This was answered well by many candidates. There were incidences of ambiguous answers, such as a 'crossed tick' (✓✗). Credit cannot be awarded if an answer is ambiguous.

(b) (i) Unambiguous structural formulae only were accepted, therefore molecular formulae were discounted and those in which the connectivity was questionable.

(ii) There were many correct answers to this question.

(iii) This proved to be one of the most challenging questions on the paper, requiring the balanced inclusion of (2)H_2O and H^+ on the left-hand side, and NH_4^+ on the right. Salts from viable reactions of specific acids were accepted.

(c) (i) Candidates who understood the $M^+ : [M+1]^+$ ratio were generally able to gain full credit here. Occasional arithmetical errors were seen.

(ii) Many candidates gave 'ester' as a correct answer but were then unable to corroborate this with sufficient and necessary information from the spectrum. Reference to an absorption outside of the C=O ester region contradicted an answer; equally, an absorption within the 1710–1750 cm^{-1} window was not solely sufficient, because of overlap with other absorptions. Corroboration could be given with reference to the C–O stretch (and the lack of O–H stretch).

C–H stretches are always ignored in these situations.

Absorptions must be accompanied by with assignment to an identified bond.

(iii) Propan-1-ol was a common incorrect answer for **T**. OH– is discouraged as a representation of an alcohol group. Many candidates did not draw the branched C5 acid correctly: some structures were unbranched; others had too many or too few H atoms.

4(a(i)	3-methylbut-1-ene	**1**

4(b)	$\frac{2.7}{48.7} \times \frac{100}{1.1} = 5(.04)$	**1**
	$M_r = 5 \times 12 + 10 \times 1 = 70$	**1**

Question 4

(a) (i) Most candidates correctly gave the systematic name of the alkene **D**, with a molecular formula of C_5H_{10}, as **3-methylbut-1-ene**.

(ii) Few candidates gave a correct explanation of why the alkenes, **C** and **D**, did not exhibit geometric isomerism.

In **C**, one C (in the C=C bond) has two methyl groups and in **D**, one C (in the C=C) has two hydrogen atoms; neither will exhibit geometrical isomerism.

(iii) Many candidates gave pent-2-ene, or **C** or **D** as examples of a positional isomer.

(iv) When **D** is reacted with $H_2(g)$, 2-methylbutane, $CH_3CH(CH_3)CH_2CH_3$, is formed. Many candidates omitted the brackets surrounding the side-chain methyl group, on the CH group.

(v) Many candidates gave a monomer containing only two methyl groups or structures that still contained a double bond.

(b) The question contained information regarding the relative intensities of the M and (M + 1) ions, from a mass spectrum of **C**, which showed a molecular ion peak at 70.

Many candidates correctly calculated that C contained 5C atoms but then failed to use this information, in conjunction with the molecular ion peak, 70, to confirm that the molecular formula of C was C_5H_{10}.

(c) (i) The mechanism for the addition of HBr to **C,** 2-methylbut-2-ene, was quite well answered. Common errors included full charges on the H-Br, instead of a dipole, curly arrows, from C=C to $H^{\delta+}$ of HBr, arrows starting at one end of the C=C bond and intermediates with both a –ve charge and a partial negative charge.

(ii) This question was well answered with a common error being the use of C_3H_7, instead of the unambiguous expanded version, $(CH_3)_2CH–$.

(iii) This question required an explanation as to why different major products were formed when HBr reacts separately with C and D.

Many candidates appreciated that methyl/alkyl groups are responsible for a positive inductive effect and that the more alkyl groups, surrounding the carbocation, the more stable is the carbocation intermediate. However, the application of this concept to the intermediates of C and D was not well recognised.

Common errors were statements that the tertiary halogenoalkane was more stable, in C, and the secondary halogenoalkane was more stable, in D, rather than their respective intermediate carbocations.

Many candidates quoted Markovnikov's rule without any further explanation.

(d) *Please note that due to an issue with diagram labelling in this question, full marks have been awarded to all candidates to make sure that no candidates were disadvantaged. This has been corrected in the published version of the paper.*

The question required completion of a table of reagents and conditions for a reaction scheme.

The formation of alcohol, **F**, from a halogenoalkane, **E**, required the use of NaOH(aq).

Cleavage of the double bond, in **C**, to form a carboxylic acid, **G**, and propanone, was carried out using hot acidified <u>concentrated</u> $KMnO_4$.

The formation of ester, **H**, from **F** and **G**, needed <u>concentrated</u> H_2SO_4 (as a catalyst).

3(a)(i)	$4P / P_4 + 5O_2 \rightarrow P_4O_{10}$	**1**
	$P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$	**1**

3(c)(i)	H_2O / steam	**1**

3(c)(iv)	C=O	**1**
	absorption changes to frequency / wavenumber / from 1670–1740 to 1710–1750 cm^{-1}	**1**

3(d)(i)	$H_3PO_4 + 3CH_3OH \rightarrow (CH_3O)_3PO + 3H_2O$	**1**

3(d)(ii)	$\frac{0.84}{12.7} \times \frac{100}{1.1} \quad (= 6.013)$	**1**
	no. of carbon atoms = 6	**1**
	molecular formula = $C_6H_{15}O_4P$	**1**

Question 3

(a) (i) Fully balanced equations were seen by more able candidates.

(ii) This definition was described accurately in most candidates answers. A common misconception was seen in weaker responses that described a weak acid as one that *'did not completely dissolve'*.

(b) (i) Many candidates gave a definition of the enthalpy change which occurs when 'one mole of product is formed from its elements'. It was less common for the elements to be described as 'in their standard states'.

(ii) More able candidates gave excellent and logical workings that contained clearly labelled energy cycles. It was common to see an expression for the different bond energies added or subtracted from each other, in a variety of possible incorrect combinations, with no attempt to show how these expressions were deduced. The stoichiometry of the equation was ignored by some.

(iii) Excellent answers were seen which identified phosphorus in H_3PO_3 as both oxidised and reduced and linked the process of oxidation and reduction to the relevant changes in oxidation number for the three phosphorus containing species. Weaker responses were less specific, for example '*H_3PO_3 is oxidised and reduced'*. Oxidation numbers were sometimes described incorrectly.

(c) (i) Some candidates identified **A** as steam or $H_2O(g)$, other candidates were less precise and identified **A** as water. A common incorrect answer was NaOH.

(ii) Many unambiguous representations of 2-bromopropane were seen. Occasionally the central carbon atom was drawn with a valency of 3.

(iii) Description of this esterification reaction as a condensation reaction was relatively well known. Incorrect answers included hydrolysis and oxidation.

(iv) This question proved to be demanding. Candidates were expected to ignore all bonds which were common to both reactant and product. While the appropriate bond, C=O, was identified in many answers C-O and C-H were also seen. Most candidates only identified a range of wavenumbers where a change in absorption occurs rather than described if the absorption appeared, disappeared or moved.

(d) (i) More able candidates deduced a formula for the correct organic product in the equation. The inclusion of H_2O as the additional product was less common, even when included the equation was not always balanced.

(ii) The appropriate formula to deduce the number of carbon atoms present in **T** was not well known and many confused calculations were seen. It was common for candidates to give a structural formula for **T** rather than the required molecular formula.

4(a(i)	3-methylbut-1-ene	**1**

4(b)	$\frac{2.7}{48.7} \times \frac{100}{1.1} = 5(.04)$	**1**
	$M_r = 5 \times 12 + 10 \times 1 = 70$	**1**

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The formation of ester, **H**, from **F** and **G**, needed concentrated H_2SO_4 (as a catalyst).

1(a)(i)	giant metallic	**1**
1(b)(i)	(the mass of) one twelfth of the mass of a C-12 / ^{12}C atom (or isotope)	**1**
1(b)(iii)	$(62.93 \times 69.15) \div 100 + (64.928 \times 30.85) \div 100 = 63.55 / 63.546$	**1**
1(c)	$^{63}Cu^{2+}$	**1**

Question 1

(a) (i) A high proportion of candidates knew that copper is metallic. Many candidates omitted that the lattice is giant.

(ii) Diagrams showing the metallic bonding as an array of positive metal ions surrounded by delocalised electrons were varied, with many either omitting or not correctly labelling the positive ions and delocalised electrons.

(b) (i) Some candidates incorrectly referred to a mole of ^{12}C atoms or just an atom of ^{12}C.

(ii) Many correct answers were seen. Some candidates omitted the word 'average'.

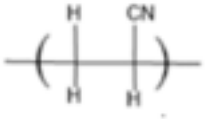
(iii) The question was quite well answered. The major error included the use of 63 and 65 as the masses of the copper isotopes instead of the data given in the question.

(c) Very few candidates recognised that the ion with an abundance of 23% and an *m/e* of 31.5 was that of $^{63}Cu^{2+}$.

(d) (i) The equation was not well answered overall. Few candidates recognised that it was a redox reaction. Many answers contained electrons on one side of the equation and/or were completely unbalanced.

(ii) The identification of Cu^{2+} as the oxidising reagent in this reaction was generally not well known. Many candidates answered with just 'copper'.

(iii) The full electronic configuration of Cu^{2+} was frequently incorrect with the $4s^2$ electrons still present in many answers.

6(a)	 OR 	**1**
6(b)(i)	nucleophilic addition	**1**
6(b)(iv)	**S** C≡N **AND T** C=C	**1**

Question 6

(a) The single repeat unit derived from the monomer, CH_2=CHCN, by addition polymerisation was quite well answered.

(b) (i) Many candidates recognised that the mechanism for step 1 was nucleophilic addition.

(ii) Overall, the mechanism for step 1 was generally not answered well. The polarity on the C=O bond was often absent and the arrow from the CN^- to the C (in the C=O bond) did not start from the lone pair on the C of the CN^- ion. Several answers contained an incorrect intermediate with a positive charge which was then attacked by the CN^- nucleophile and many answers incorrectly showed an arrow from an H^+ ion to an O^- (in the intermediate).

(iii) Some good answers were seen.

(iv) The absorptions in the infrared spectrum of CH_2=CHCN were often correctly identified as C≡N and C=C. Incorrect answers mainly contained an identification of the functional groups rather than the bond or identifying the absorption (annotated as T on the IR spectrum) as that of a C=O group.

(c) (i) Very few candidates gave an accurate description.

(ii) A number of answers were awarded some credit. Few candidates were able to give a full description.

(d) Several candidates gave a correct pair of molecules that were non-superimposable images of $CH_3CH(OH)CN$. A common error was diagrams containing the same type of bond at 180° to each other.

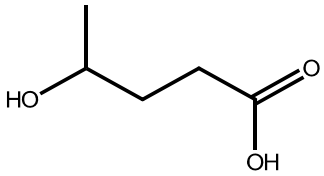
(e) The equation for the reduction of propanenitrile to propylamine using hydrogen gas was not well known. Few candidates used the correct stoichiometry for hydrogen.

(f) (i) It was well known that the name of the type of reaction in step 1 is substitution.

(ii) Many candidates gained credit for ammonia. Many omitted heat or the solvent as conditions.

5(a)	molecules / isomers with the same molecular formula / same number of atoms of each element **AND** different structural formulae / different structures	**1**

5(d)(i)	ester	**1**

5(d)(iii)		**1**

5(d)(iv) Any two of the following differences in spectra **D** compared to **E**: **2**

	bond	**functional group**	**value or range of values of absorbance inclusive (cm^{-1})**
E	*C=O / C double bond to O*	*(ester)*	*1710–1750*
D	*C=O*	*(carboxy(l))*	*1670–1740*
D	*O-H OR OH bond*	*(carboxy(l))*	*2500–3000*
D	*O-H OR OH bond*	*(hydroxy(l) OR alcohol)*	*3200–3600*

Question 5

(a) This definition was generally well known.

(b) (i) Many candidates appreciated that the two molecules were members of the carbonyl functional group. Some candidates incorrectly referred to them as either aldehydes or ketones.

(ii) Correct structures of **A** were common. Many candidates represented **B** as a ketone; the correct carbon skeleton was not always deduced.

(c) (i) It was common for some of the information in the question to be used to produce partially correct structures. Many candidates incorrectly produced structures with two types of stereoisomerism.

(ii) This was generally well answered.

(d) (i) Ketone and carboxylic acid were the most common incorrect answers.

(ii) Hydrolysis was a common incorrect response.

(iii) Deduction of the structure of **D** proved to be demanding. Often the structure drawn had a carboxylic acid group at one end of the molecule; identification of $(CH_3)CH(OH)$– at one end of the molecule was seen less frequently. Occasionally, the correct ends of the molecule were drawn with an incorrect number of carbon atoms in the carbon backbone.

(iv) Stronger candidates clearly identified two differences between the spectra in terms of the different absorption ranges produced as a result of the different environment of the C=O bond in an ester compared to a carboxyl group, or in terms of the presence of the O-H bond in a hydroxyl or carboxyl group which is absent in the ester, **E**. Reference to absorbance at 2850 – 2950 due to C-H bonds was common, even though both **D** and **E** contain this type of bond.

3(a)	**M1** *appearance* (colourless / yellow) liquid **M2** *structure and bonding* simple / molecular **AND** covalent	**2**

3(f)	**M1** *m/e 164* $(CF_2{}^{35}Cl^{79}Br)^+$ **M2** *m/e 166* $(CF_2{}^{37}Cl^{79}Br)^+$ **AND** $(CF_2{}^{35}Cl^{81}Br)^+$	**2**

Question 3

(a) Very few candidates could state the appearance of $SiCl_4$ at room temperature; more were able to state the correct structure and bonding. The most common incorrect response was giant covalent.

(b) Most candidates correctly referred to the appearance and pH. Candidates often did not refer to the relevant observation and struggled to differentiate the behaviours of the two chlorides.

(c) (i) A large number of candidates stated the correct reagent. Many left out aqueous or dilute as the conditions.

(ii) Many of the candidates defined disproportionation; a small proportion were able to refer to the correct oxidation numbers. Several referred incorrectly to the +1 oxidation number for chlorine.

(d) (i) This was well answered by most candidates.

(ii) Most candidates correctly answered the initiation step. Many candidates found the propagation and termination steps more challenging.

(e) This was well answered by most students.

(f) Many candidates were able to give the correct formulae at m/e = 164.Very few candidates gave both of the correct options at m/e = 166.

4(a)(i)	NaOH in alcohol / ethanol AND heat (under reflux)	**1**

4(d)(i)	**G**	**1**
	(absorption at) 2200–2250 (cm^{-1}) **AND** C≡N	**1**

4(d)(ii)	3.4 **AND** relative abundance of ^{79}Br:^{81}Br ≅ 50:50 OR 1:1	**1**

Question 4

This was a second organic chemistry question, which included some analytical work. This was generally answered well by many candidates, though later parts on IR and MS appeared less familiar to some.

(a) (i) 'Heat' or equivalent was often missing. H_2SO_4 was a popular answer despite the conditions suggested in the syllabus.

(ii) Many repeat units were shown containing four carbon atoms, as if the CH_3 groups had been incorporated into the backbone. Some candidates showed an ethene structure with missing CH_3 side groups.

(iii) Candidates very often showed the reverse equation with C_4H_8 and H_2O. Quite a few answers contained Al_2O_3 as a reactant and several answers omitted H_2O as a product.

(b) (i) This was well answered.

(ii) There were many correct structures/names for iodoform but very few correct propanoate ions.

(c) (i) There were many correct answers to this item.

(ii) Catalyst and oxidising agent were the common errors here, but otherwise there were many correct answers.

(d) (i) Compound **F** was commonly given incorrectly as the unknown compound, referring to an O–H absorption in the range 2850–3000 cm^{-1}. Candidates are reminded that C–H stretches appear in all organic spectra. Where absorptions are referenced, candidates need to identify which bond is stretching: 'nitrile', for example, is insufficient.

(ii) There were many null responses to this. Where attempted, candidates' answers were evenly split between correct identification of an identical abundance and some complicated (erroneous) mathematics.